

L'Analisi dei Requisiti nel Processo di Usability Engineering

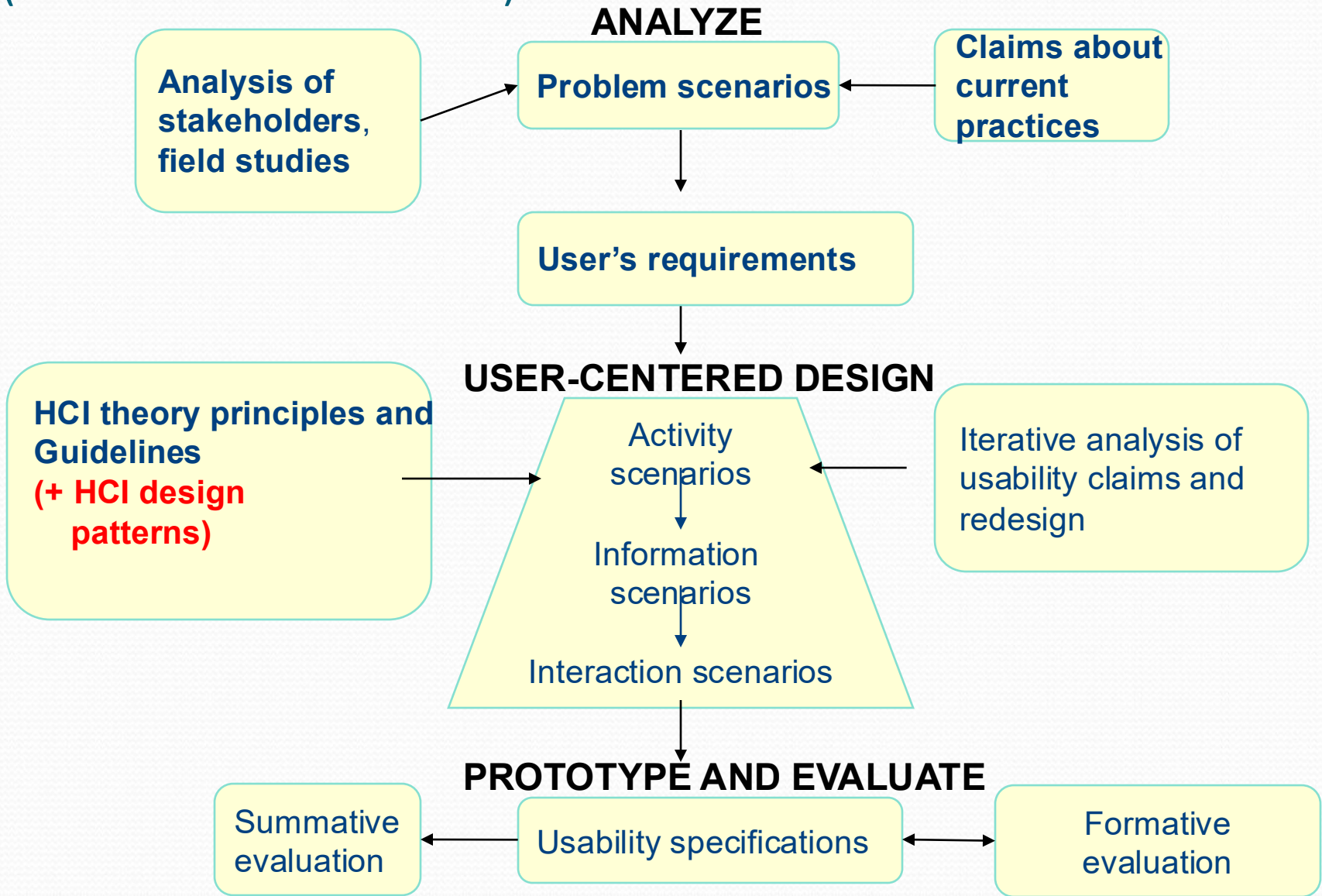
Corso di Informatica – Laurea Magistrale

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A scenario-based design process

(Rosson and Carroll 2002)



Scenarios as a support to solution-first problem solving strategy

Hazards of the Solution-First Approach

- Designers want to select a solution approach quickly, which may lead to premature commitment to their first design ideas.
 - Designers attempt to quickly simplify the problem space with external constraints, such as the reusing of familiar solutions.
 - Designers are intent on elaborating their current design proposal, resulting in inadequate analysis of other ideas or alternatives.
- How Scenario-Based Design Can Help
 - Because they are concrete but rough, scenarios support visible progress, but also relax commitment to the ideas expressed in the scenarios.
 - Because they emphasize people and their experiences, scenarios direct attention to the use-appropriateness of design ideas.
 - Because they are evocative and by nature incomplete, scenarios promote empathy and raise usage questions at many stages



- A scenario is a description of a person's (or more people's) interaction with a system.
- Scenarios help focus design efforts on the user's requirements, which are distinct from technical or business requirements.
- Scenarios may be related to 'use cases', which describe interactions at a technical level. Unlike use cases, however, scenarios can be understood by people who do not have any technical background. They are therefore suitable for use during participatory design activities.

The initial step – an ethnographic study

- Ethnographic study is ‘an approach to inquiry that focuses on culture or subculture of a group of people, with an effort to understand the world view of those under study’ [Polit et al., 1995]
- Since early 90’s researchers in HCI/CSCW have focused on the adoption of ethnographic studies during software requirements and design processes. [Hughes et al., 1992, Sommerville et al., 1993]
- Interviews and surveys are a way to get information from people and their results can be used to study in terms of how humans think, do and what they prefer.
 - focus on the ways in which such studies may be organized and presented as part of the requirements process or within design or project meetings

Contextual inquiry and observation

Goal

- to identify requirements within context of use

Need to

- take account of stakeholders
- study typical organizational contexts with relevant business process activities
- envisage suitable scenarios of use

Analyzing Users' Requirements

Goal: understand users' current activities well enough to reason about technology-based enhancements

- Understanding the work being done now
 - so as to offer function that meets real needs
- As well as learning about the people themselves
 - so as to offer function in a way that is convenient and satisfying

Analyzing Work

- Observe and describe people's activities
 - what goals do they pursue, how?
- Collect and study artifacts used in these activities
 - tools, documents, features of the work setting
- Capture the social context of the work
 - groups and organizations, roles and relationships

Examining an Artifact

- What does it tell you about the task it supports?
 - if at all possible, observe it *in use*
 - objects are not always used as intended :-)
- Try to extract task information and procedures
 - what task attributes are apparent or can be inferred?
 - what action sequences are required or possible?
 - what seems likely to be simple or difficult to do?
- Practice on some familiar examples:
 - ex: appointment book, wristwatch, badge

Analisi di un artefatto

- Esaminare un'agenda attualmente usata da voi o da una vostra conoscenza. Cosa vi dice sui task di scheduling e di ricerca di eventi? Che informazioni specifiche vi dà circa l'individuo/gli individui che la usano?
- Pianificare e cercare un appuntamento:
 - alcuni scheduling coinvolgono persone, altri argomenti, altri entrambe le cose
 - può essere citato un posto
 - possono essere elencati i partecipanti a un meeting
 - può essere annotata la durata
 - il tempo potrebbe non combaciare con le ore stampate nell'agenda
- Circa l'utente:
 - Questa persona usa l'agenda per le cose da fare e per gli appuntamenti
 - È una persona impegnata!
 - Questa persona non sempre scrive appuntamenti che si ripetono

Getting Users Involved

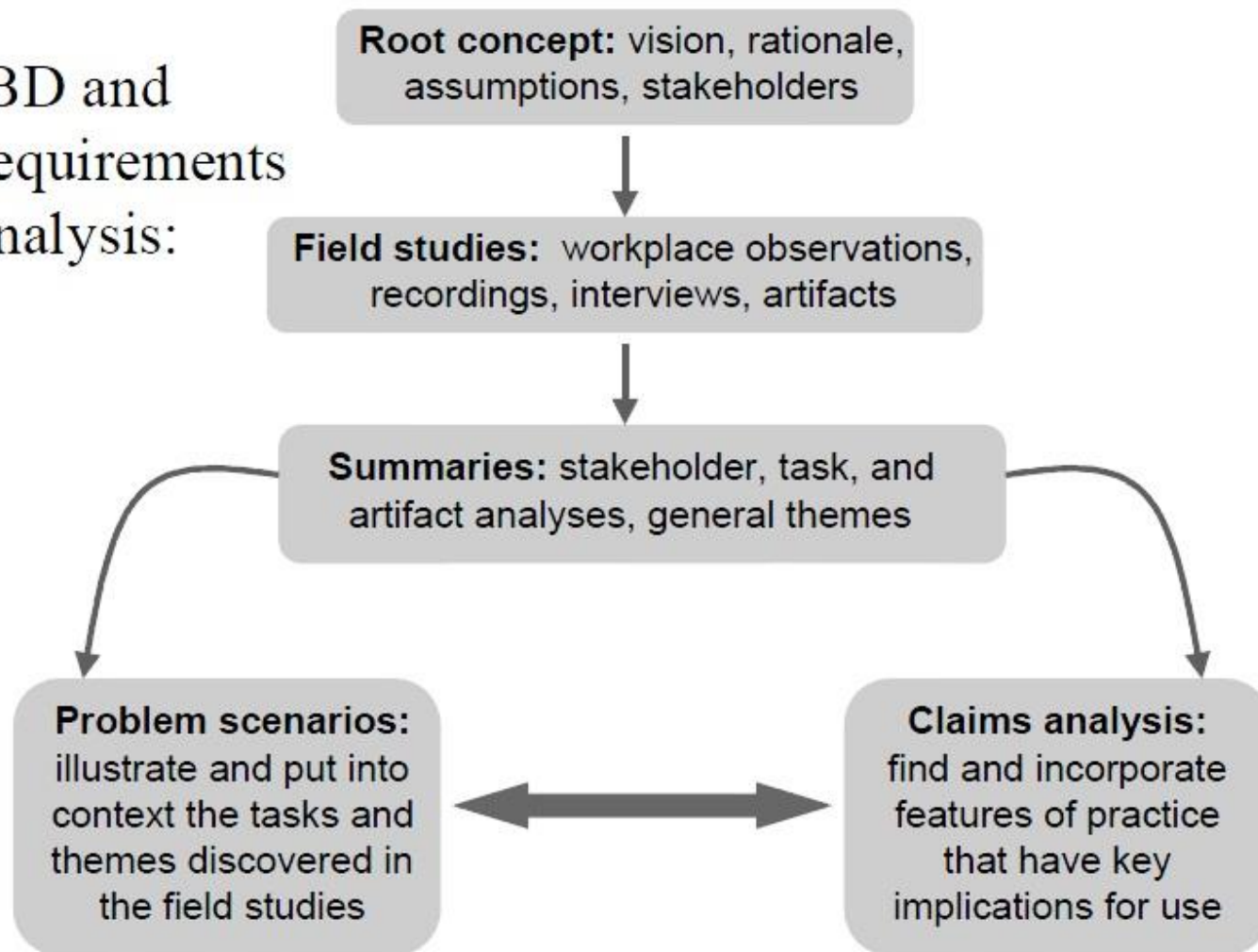
- Usually there will be multiple “stakeholders”
 - e.g., workers, but also support staff, management
 - each with knowledge, preferences, perspectives
- Observe and/or interview representatives from all relevant groups
 - discuss their typical tasks, their role in the organization
 - as well as technology background and expectations
- *Participatory analysis: videotapes or other records of activities that participants view and discuss*

People Know More than They Can Say

- Trouble Ticketing System (TTS)
 - database for tracking telephone line problems
 - relieve need for personal follow-up by staff members
- But, personal follow-up played a key role
 - details of problem filled in, organizational knowledge
 - TTS led to ‘workarounds’, *with phone contact made as usual*, but TTS used just to document jobs later
- *Tacit knowledge: engineers were not aware how critical this social exchange was to their jobs*

Le prime fasi dello Scenario-Based Approach

SBD and
Requirements
Analysis:



Le prime fasi dello Scenario-Based Approach

- Organize as a group, develop root concept
 - Observe activities, gather artifacts (at store)
 - Interview a representative number of stakeholders
- Create summary diagrams and tables
- Express understanding in problem scenarios
 - develop hypothetical stakeholders to use as actors
 - integrate concerns and opportunities discovered in a believable usage context
- Extend scenario analysis with claims

Capitalizing the acquired knowledge - Problem Scenarios

- Describing *personas* as prototypical stakeholders
 - biography
 - motivation
 - goals
- Depicting scenarios of established work practices, where *personas* are engaged, from which user's requirements will be derived

Capitalizing the acquired knowledge - Claims

- Designer's understanding of the current situation is communicated in
 - problem scenarios and
 - claims

Problem Scenarios

- A problem scenario tells a story of current practice. These stories are carefully developed to reveal aspects of the stakeholders and their activities that have implications for design.

Scenario 1: Current Stanford University Learning Design and Technology (LDT) program student

Actor: Jane joined the LDT master's program at Stanford in June. She had worked in an educational setting for a few years before joining LDT. She is very interested in designing technology for educational purposes. She still hasn't figured out her career plans upon graduation, but hopes that she can be involved in a career that designs educational software and encourages creativity. Right now, she is in the midst of searching for ideas for her master's project.

Background: BSc degree, very comfortable with computers, Microsoft office software, intermediate skills on Web publishing tools (e.g. Macromedia DreamweaverTM) and graphics software (e.g. Adobe PhotoshopTM)

Scenario 1: Current Stanford University Learning Design and Technology (LDT) program student

Jane is required to post her master's project proposal and she is looking for ideas. She surfed the Web and was hoping to find some inspiration. But she found herself spending hours doing research. She was not very productive since the information she was looking for was scattered on many different Web sites. She thinks it'll be really useful to her if there is one site that will provide information on all of the design.

Claims

- Claims enumerate features of the current situation that are understood to have important consequences – both positive and negative-to the scenario actors

Claims about Scenario 1

Situation Features	Pros (+) and Cons (-)
Trying to gather design ideas and inspiration on the Web	+ Web is a good resource for exploration and researching + It is a convenient way to do research - Can be easily overwhelmed by the vast resource - It takes time to sift through the relevant information - It requires some skills to effectively search the Web
Trying to search for information about expert in this field from the Web	+ Web is a good resource and convenient way to look for information - Many experts don't have their own websites - Very unlikely to find information about the design approaches, experience etc. of experts on the Web since this information is usually not published
A library that summarizes different design approaches	+ Good for browsing and grasping the key concepts + Ease for comparing and contrasting between different design approaches + Ease of reference - Takes time and effort to gather relevant information and create a summary

Scenario 2: Graduate of LDT currently working at a design firm

Actor: Larry N. is an experienced multimedia designer working for a top software firm and is charged with designing marketing products that seed the education market. He designs tutorials or other media that help people get started with the software product. He has been working for this company for three years and prior to that he worked for an educational software company targeting the K-5 literature audience. Altogether he has about six years working with software in an educational setting.

Background: His background includes four years as a layout artist and three years as an elementary school teacher. His BA is in Philosophy with a minor in art. He is thirty-six years old and is self-taught in Web design and action script programming. He does not belong to any professional organizations, but does subscribe to some online design newsletters and has a subscription to an action-script programming newsletter. He tries to attend at least one workshop on the new features of the software and other types of professional conferences/tradeshows once a year.

Scenario 2: Graduate of LDT currently working at a design firm

Larry has to revise an existing product that was designed two years ago. He was the junior designer in a two-person team. The senior member has left the team. Larry remembers the product very well because occasionally he gets a phone call from the sales force people to ask about a couple of the features of the product. He desires to get a new frame of reference to approach the redesign. Normally, he and his teammate would just sit down and review the product, maybe take in some feedback from the sales people and his boss and begin the redesign. Because this is a redesign he is expected to create a new product within six weeks. He gives himself 30 minutes to do some research.

Claims about Scenario 1

Situation Features	Pros (+) and Cons (-)
Trying to gather redesign ideas from the Web	+ Web is a good resource for exploration and researching + It is a convenient way to do research - Can be easily overwhelmed by the vast resource - It takes time to sift through the relevant information - It requires some skills to effectively search the Web - Very unlikely to find information about the design approaches, experience etc. of experts on the Web since this information is usually not published
A library that summarizes different design approaches	+ Good for browsing and grasping the key concepts + Ease for comparing and contrasting between different design approaches + Ease of reference - Takes time and effort to gather relevant information and create a summary

Scenario 3: Professors of LDT

Actors: Professors Hoadley and Walker, two experienced education design professors visiting a designer interview web site. Their aim may be to find innovative solutions to design problems they have encountered, find designers who they believe are luminary in their innovation or their unique approaches to design problems, and to search for possible projects that they could impose on the next years poor students.

Background: In addition they have been in the field for some time now and have developed a knowledge of the key designers for education, as well as the sites around the world that have consistently provided reliably good education design models.

Scenario 3: Professors of LDT

Time is limited for these busy professors so an indexing system that either categorizes the work by approach or by influences would be preferable. With respect to time, they would prefer that you summarize the designer's theoretical model and link these with a searchable key word. In this way designers working within the same theories could be compared. The search should exclude unrelated design fields and nonacademic institutions.

As opposed to the main users for this site who are the LDT students, these professors are able to quickly scan large amounts of data from the designer of interest to find the relevant information. Furthermore in depth bibliography of the designers work would have to be easily referenced so that they could study their target designer in depth. The average LDT student would have little time for this information.

Claims about Scenario 3

Situation Features	Pros (+) and Cons (-)
Pictures of designers	+ Easy to recognize - Download times increase
Academic designers	+ Tend to fit more readily into predefined theories and are more theoretical than practical - Interests of the LDT group is more varied
Include designers influences	+ Insight into their work without having to read the full interview - Time consuming to collect all this data
Educational design teaching tools used analysis	+ Help students model teaching projects around these ideas - Little interest in teaching educational design among LDT group
Links to Bibliography	+ More in depth analysis possible + Ease of reference - Time consuming and not particularly educational or helpful for LDT group

A case study...

- We were tasked with improving a company's business processes through a system that allows for the sharing of spreadsheets.
- A contextual investigation revealed that spreadsheets are widely used by small businesses, but predominantly with a single-user system that allows for the sharing of printed sheets.
- Business processes are slow and errors are frequent. The need for technology that supports sharing is recognized, but...
- ...there's no room for significant investment, even in terms of staff training costs

The fieldwork

The 5 companies:

- boat cruises
- interiors supply
- building construction
- financial services
- no-profit company, set up to develop and deliver training programs for unemployed people.

THE MAIN GOAL OF OUR OBSERVATION

What obstacles in their current setting hinder the full exploitation of the powerful functionalities of spreadsheet applications?

The fieldwork results

From the observation, it turned out that:

- Spreadsheets are intensively used for crucial management activities by almost all the interviewees.
- However, in most cases, collaboration happens by means of written annotations on spreadsheets printouts, with very few exceptions in the use of on-line spreadsheets, provided for reading access only.

We enquired the interviewees about

- the work activities they usually perform with spreadsheets and
- problems occurring with the physical, rather than digital, sharing of such artifacts, and
- what they would expect from a web-based collaboration through spreadsheets, if a system supporting it could be deployed inside their organization.

The fieldwork findings

The major findings of this initial fieldwork:

- collaboration through spreadsheet is recognized to be a powerful means to enhance crucial activities carried out in SMEs.
However,
- the main obstacles to its realization appears to be the small size and the limited human and financial resources, which usually characterize SMEs themselves.
- the first general requirements are therefore:
 - Low training cost – the overall enhancement of the business process should be gained with little learning effort
 - No maintenance costs – companies should be able to easily maintain the system, once it is deployed
- further requirements, closely related to collaboration, came from the observed activities.

Capitalizing the acquired knowledge

PERSONA 1

Mark Ross is the manager and the owner of a staircase manufacturing company. He has a long experience with this job, having started 20 years ago as a staircase worker in a bigger company. 5 years ago he has set up his own company and has currently 20 employees. He uses spreadsheets for scheduling resources and activities related to each job commitment and for periodical statistical analysis required for accounting, payroll and retail activities performed through the Australian MYOB business software package. He is aware that several benefits could come from the on-line sharing of spreadsheets. However, he is afraid that the training activities that employees should undergo to actuate such transformation would be too costly and even ineffective for those employees who have limited familiarity with computer systems.

BIOGRAPHY

MOTIVATION

GOAL

Capitalizing the acquired knowledge

Scenario 1 –

The two ladies from the front desk

- keep records of customers' inquiries and commitments in a unique diary, which is daily passed over to Mark.
- have read-only access of the spreadsheet file with commitments details

Mark

- updates the spreadsheet file that the two ladies access during their work, and
- saves a copy of that file in his personal laptop computer.

When a commitment is received,

- creates a new spreadsheet, with pricing details and customer's requirements
- schedules the activities for that job and
- sends a printed copy of the spreadsheet to the foreman and another one to the floor manager.

Capitalizing the acquired knowledge

Scenario 1 (*cont.*) –

The foreman

- manages the allocation of workers making written annotations on the received printout and
- transmits a (paper) copy of the annotated spreadsheet to Mark

The floor manager

- accesses and modifies the stock as required and
- if needed, places new orders of building materials.
- he annotates his copy of the spreadsheet and sends it back to Mark

Mark

- properly inserts the hand-written annotations made on the distributed copies to keep the spreadsheet file updated.

Claims

Situation Features	Pros (+) and Cons (-)
Mark is the only user with read/write privileges.	+ Preservation of existing privacy and data access rules + Appropriate synchronization policy among users who have access privileges for the same portions of a spreadsheet - Possible human error Mark may misunderstand written annotation from his coworkers - Flows due to a delayed (daily) updating of the spreadsheet file (e.g., the foreman allocates 8 working hours to worker A for Tuesday 25 Aug and, upon receiving a new spreadsheet from Mark, he accesses the digital spreadsheet file and reallocates the same worker for the same day: the previous updating was not recorded yet by Mark).

Capitalizing the acquired knowledge

PERSONA 2

BIOGRAPHY

Fanny Price is a full-time experienced officer belonging to the administration staff of a community service centre. She works on training services and, often in cooperation with 3 colleagues, who work on different business units and at different locations, she manages customers' re planning activities, budgeting, wage records and so forth. She makes use of spreadsheets and often shares them by e-mail with colleagues and with two supervisors, a General Manager, who performs cash flow analyses and periodically reports results to the Board, and a Chief Executive Director, who is responsible for the overall community development.

MOTIVATION

Fanny often experiences version consistency problems with the spreadsheet files. She believes that great benefits would come from a synchronized sharing of spreadsheet data.

GOAL

Capitalizing the acquired knowledge

Scenario 2 –

Fanny has to organize the closing ceremony of an evening training course held for immigrants.

Fanny

- consults her copy of the financial spreadsheet to verify the amount currently allocated to her unit, focusing on the cells related to social events expenses
- formulates a budget scheduling for the event and sends an email to the general manager.

General Manager

- approves the event and properly updates the financial spreadsheet, distributing the updated version to all the units.

Capitalizing the acquired knowledge

Scenario 2 (*cont.*) –

Fanny

- upon approval, consults the conference room agenda, which is again a spreadsheet file shared by all the business units, by e-mail transmission
- selects the date for the ceremony, indicates the details concerning catering expenses and number of people expected, and notifies them to the colleague from the Function and Conference Unit
- **The Function and Conference Unit officer**
- replies that earlier in the morning she has received a request for the same date from the Development Office

Fanny

- chooses another date and notifies it to the unit.

The Function and Conference Unit officer

- is now able to update the spreadsheet file accordingly.

Claims

Situation Features	Pros (+) and Cons (-)
Writing privileges are allowed to one person/unit for each spreadsheet and reading privileges are allowed to all the co-workers, but limited to the cells of interest	+ The separation of concerns relieves employees from unnecessary responsibilities - Human error: Fanny may fail in opening the last version of the spreadsheet saved in her PC. - Colleagues are not aware of <i>who</i> modified <i>what</i> in a shared artefact. Versioning control and detailed modification loggings should be devised at cell level.
One person manages financial requests	+ This is a convenient way to manage concurrent requests -Conflicting requests may derive from a delayed delivery of the updated financial spreadsheet e.g. Fanny allocates 2500 AUD for social expenses , thus leaving an amount of 300 AUD in the corresponding cell. The officer from the Development Unit makes soon after his own request, and is not aware of the last modification until he receives the updated file from the General Manager.

Let's go back to exercise on the Object-based model

- Synthesizing objects and actions
 - the *what* and *how* of users' tasks
 - objects come directly from users' tasks
 - e.g. articles, words/phrases, passages, sections
 - e.g. tracks, albums, length
 - Properties of objects
 - distinguish instances of objects
 - Actions are methods the users use to do the tasks
 - e.g. open/read, close/write, modify, use, delete

Objects

- Kiosk/Check-in desk
- Passport
- Flight booking
- Seat map
- Baggage
- Boarding pass
- Airline System

Actions

- *initiate check-in*
- *present, scan*
- *retrieve booking details*
- *select/change seat*
- *check baggage*
- *print, download*
- *display booking details, confirm ...*

Object-Action Sequence

- User → Kiosk: initiate check-in
- User → Passport: scan
- Kiosk → Flight booking: retrieve reservation
- Airline System → User: display reservation details
- User → Seat map: select or confirm seat
- User → Baggage: Place baggage for tagging and weighing
- Airline System → Printer: Print baggage tags and boarding pass
- System → Booking: Update status to *checked in*.

Problem Scenario for the Flight Check-in

It's your turn ...

Depict the scenario of a traditional flight check-in process...